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*Knowledge compounds
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High quality notes for
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CONCEPT CLARITY

Simplified explanations
with easy examples



WELL STRUCTURED

Organized content for
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TRUSTED BY STUDENTS

Built for learners,
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Mechanical Engineering Complete Guide.

Jobs, Salaries etc.

PREPARED BY

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Mechanical Engineering: Complete Guide

1. What is Mechanical Engineering?

Mechanical engineering deals with machines, engines, vehicles, energy systems, manufacturing, robotics, thermal systems, and industrial equipment. It is one of the broadest engineering fields.

2. Types of Work in Mechanical Engineering

Designing Machines:

Mechanical engineers design automobiles, engines, turbines, robots, machine components, and manufacturing systems using tools like AutoCAD, SolidWorks, CATIA, and Fusion360.

Manufacturing & Production:

They supervise production processes, manage machine operations, and ensure product quality in industries.

Machine Maintenance:

Mechanical engineers maintain machinery in power plants, factories, oil & gas plants, and manufacturing units.

Thermal Engineering:

They work with boilers, heat exchangers, HVAC systems, refrigeration, and power generation.

Automobile Engineering:

Roles include development of engines, EV systems, vehicle design, and testing.

Aerospace & Defence:

Mechanical engineers contribute to rocket systems, aircraft, missiles, and propulsion units.

Robotics & Automation:

They help build robots, automation systems, CNC machines, and mechatronics equipment.

3. Job Positions After Mechanical Engineering

- Mechanical Design Engineer
- Production Engineer
- Maintenance Engineer
- Thermal Engineer
- Automobile Engineer
- Aerospace Engineer
- Quality Control Engineer
- Project Engineer
- R&D; Engineer



4. Salary After Mechanical Engineering (India)

- Freshers: ■2.5 LPA – ■6 LPA (normal), ■8 LPA – ■14 LPA (top companies)
- Mid-level: ■8 LPA – ■18 LPA
- Senior level: ■15 LPA – ■40 LPA+
- Abroad: ■25 LPA – ■80 LPA+

5. Top Companies for Mechanical Engineers

Automobile:

Tata Motors, Mahindra, Maruti Suzuki, Hyundai, Hero MotoCorp, Ashok Leyland.

Core Mechanical:

L&T, Thermax, Kirloskar, Bosch, Siemens, Godrej Aerospace.

Oil & Gas / Energy:

ONGC, IOCL, Reliance Industries, BPCL, HPCL.

Aerospace & Defence:

ISRO, DRDO, HAL, Boeing, Airbus.

Manufacturing MNCs:

Mitsubishi, GE, Schneider Electric, ABB, Caterpillar.

Power & Heavy Machinery:

BHEL, NTPC, Adani Power, JSW Steel.

6. Is Mechanical Engineering a Good Career?

Yes—if you develop skills like CAD designing, ANSYS, CNC, automation, Python, EV technology, and project management. Skills determine salary and placement opportunities.

7. Success Roadmap (Summary)

- Master CAD tools (AutoCAD, SolidWorks)
- Learn analysis tools (ANSYS)
- Gain manufacturing knowledge
- Work on real projects

- Do internships
- Improve communication and presentation skills
- Explore EV, robotics, automation fields.

